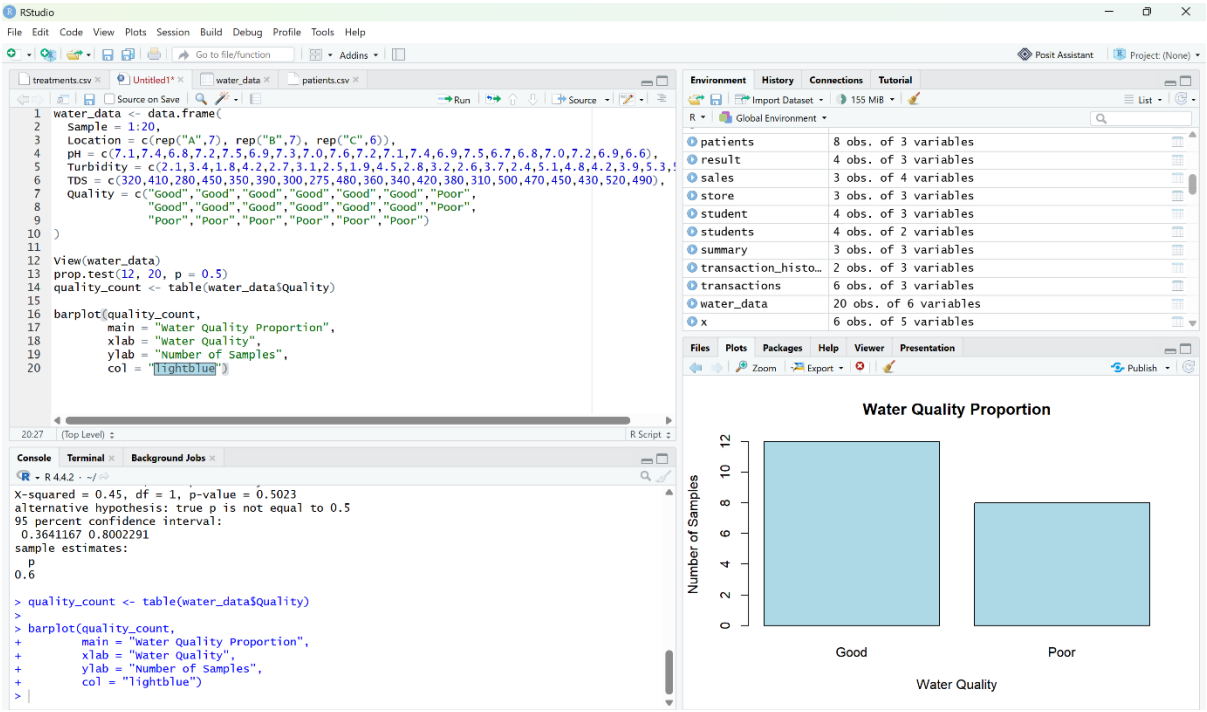
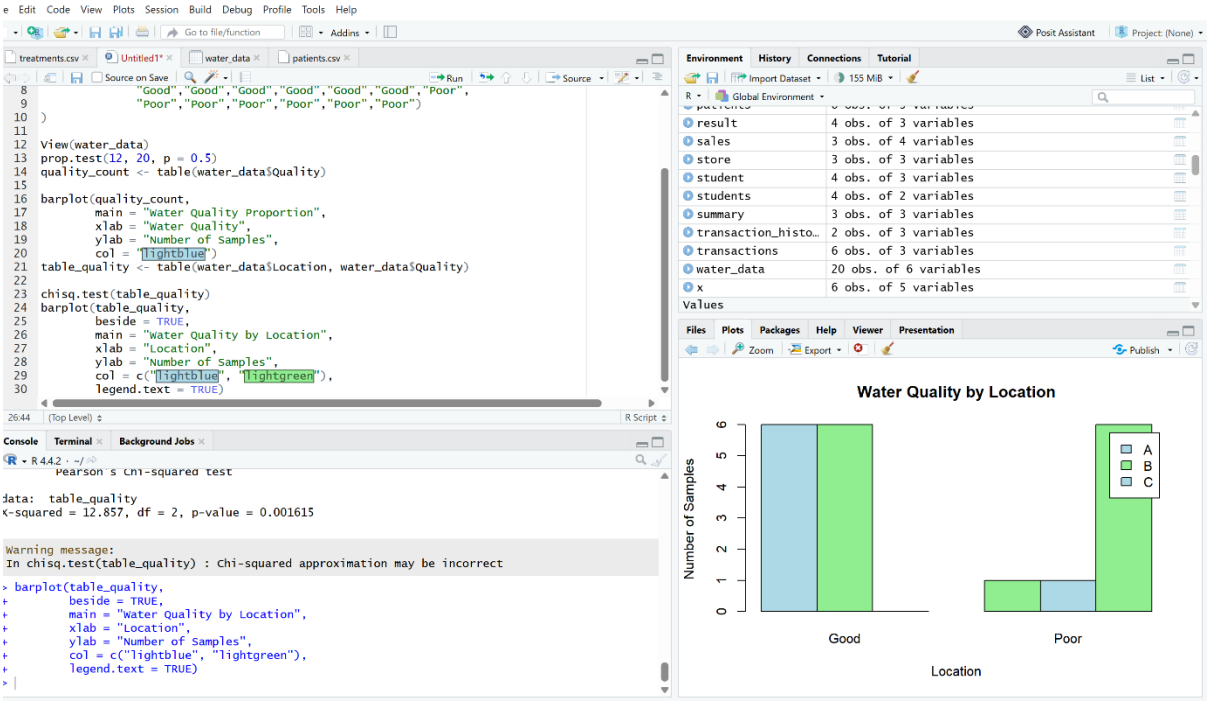
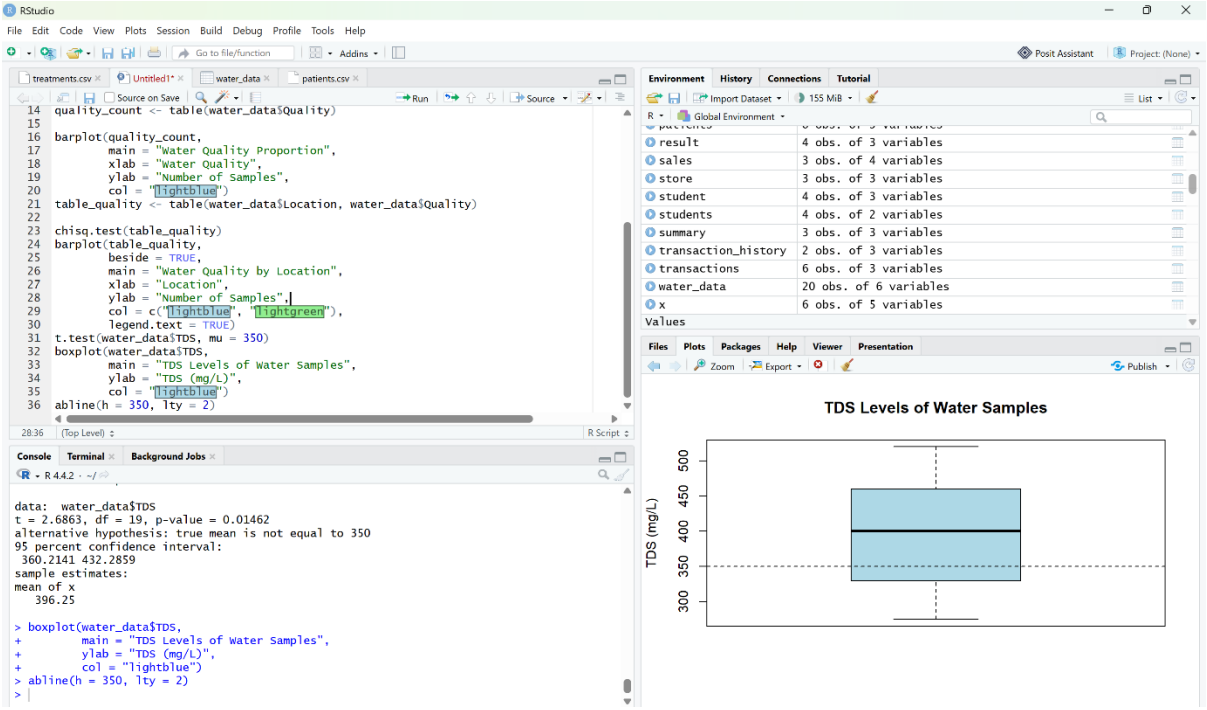
Question – 1



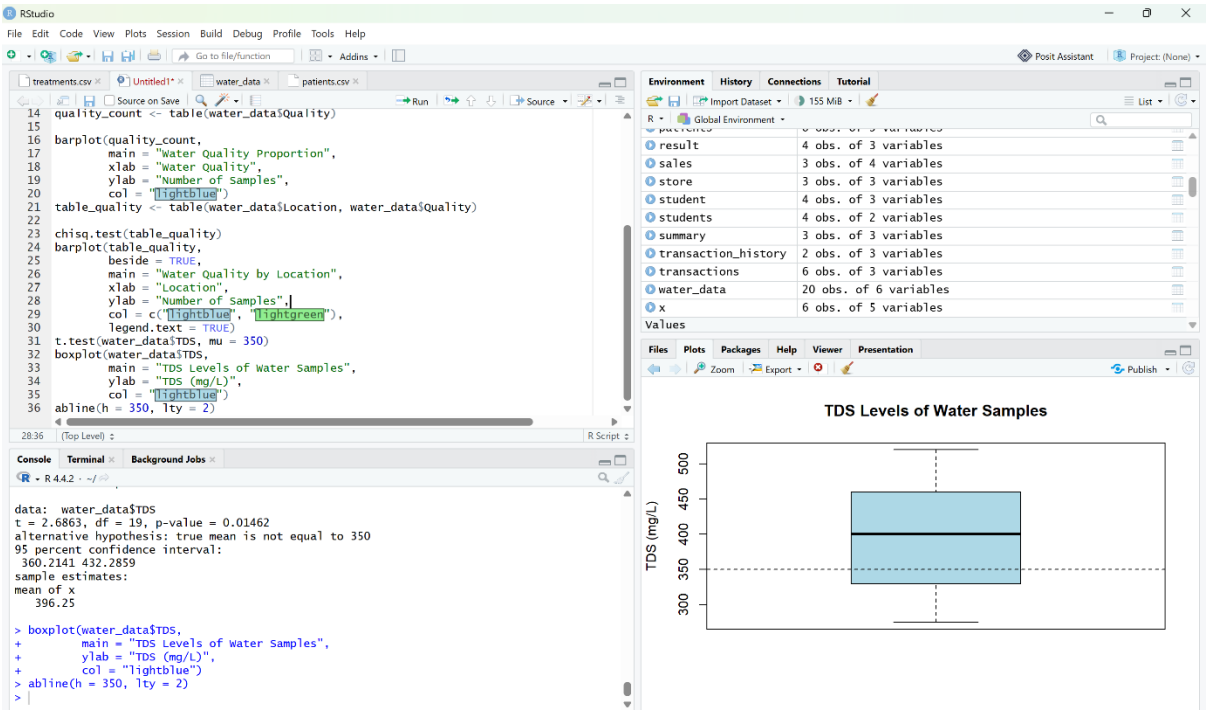
Question - 2



Question - 3



Question - 4



Question - 5

```
boxplot(water_data$TDS,
        main = "TDS Levels of Water Samples",
        ylab = "TDS (mg/L)",
        col = "lightblue")
abline(h = 350, lty = 2)
anova_result <- aov(TDS ~ Location, data = water_data)

summary(anova_result)
boxplot(TDS ~ Location,
        data = water_data,
        main = "TDS Levels by Location",
        xlab = "Location",
        ylab = "TDS (mg/L)",
        col = "lightblue")
cor.test(water_data$pH, water_data$Turbidity)
plot(water_data$pH,
     water_data$Turbidity,
     main = "pH vs Turbidity",
     xlab = "pH",
     ylab = "Turbidity (NTU)",
     pch = 19)

abline(lm(Turbidity ~ pH, data = water_data))
```

```
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.69559532  0.09174785
sample estimates:
       cor
-0.3656171

> plot(water_data$pH,
+      water_data$Turbidity,
+      main = "pH vs Turbidity",
+      xlab = "pH",
+      ylab = "Turbidity (NTU)",
+      pch = 19)
>
> abline(lm(Turbidity ~ pH, data = water_data))
>
```

